Internet of Things Fundamentals

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Semester	6th
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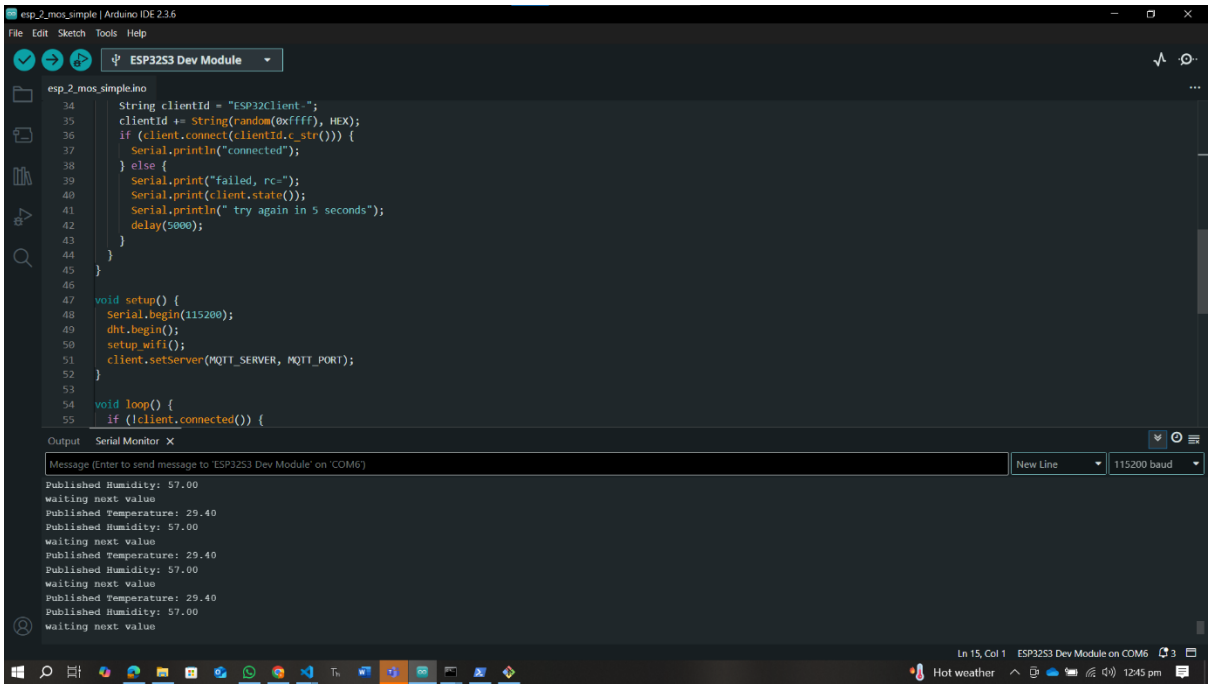
Title:

Lab 13

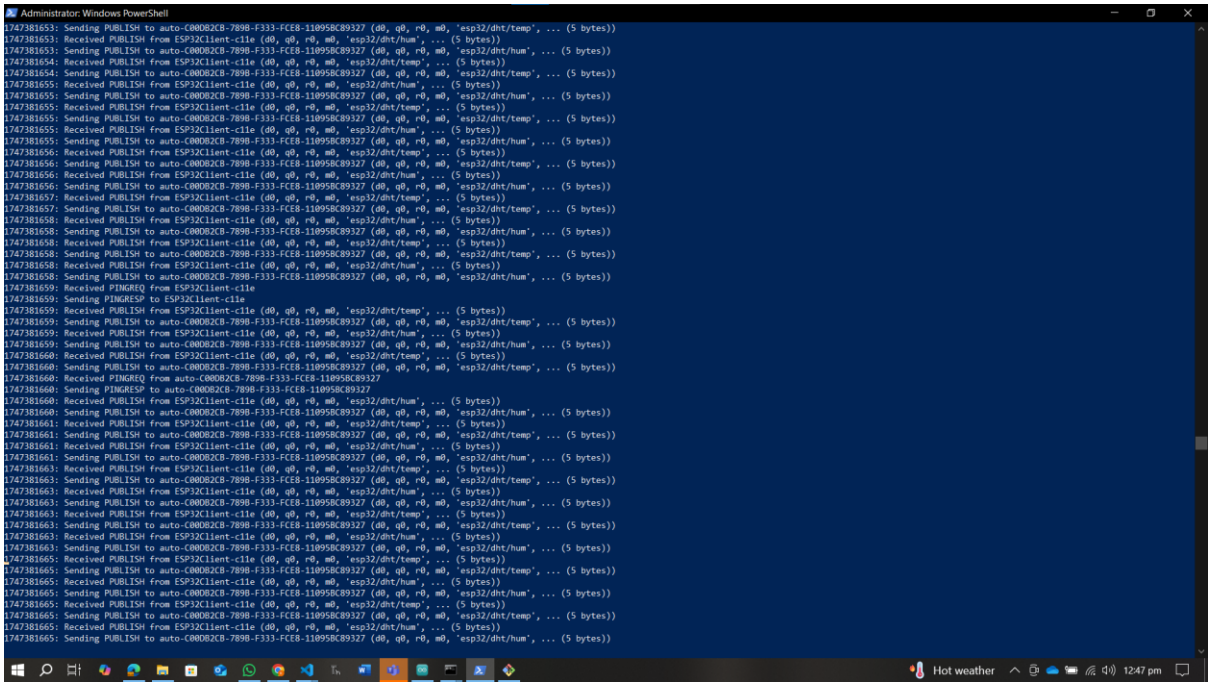
Task 1:

Run the Arduino-based code to **publish DHT sensor data to the Mosquitto MQTT broker**.

Here I run the Arduino code to get data from sensors and publish that data to MQTT broker. The ESP32 connects to Wi-Fi and publishes data every second. The serial monitor is showing live updates.



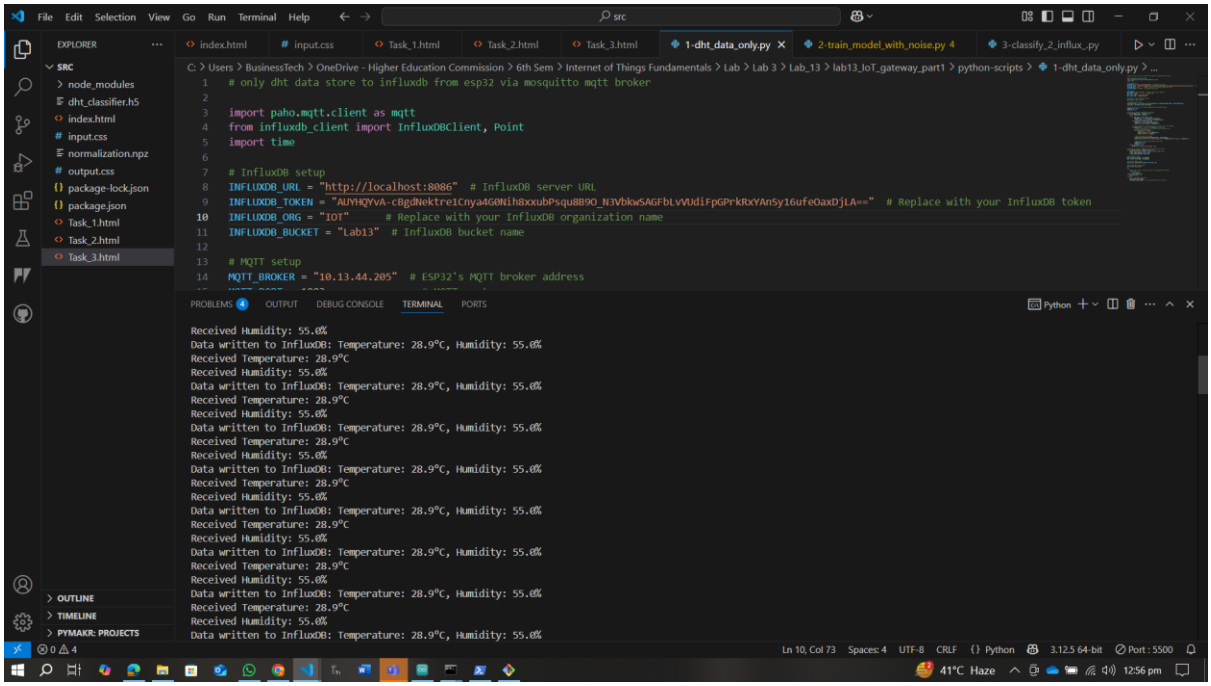
Here on the PowerShell, I check that my Moquitto MQTT broker is running successfully and get the real time data.



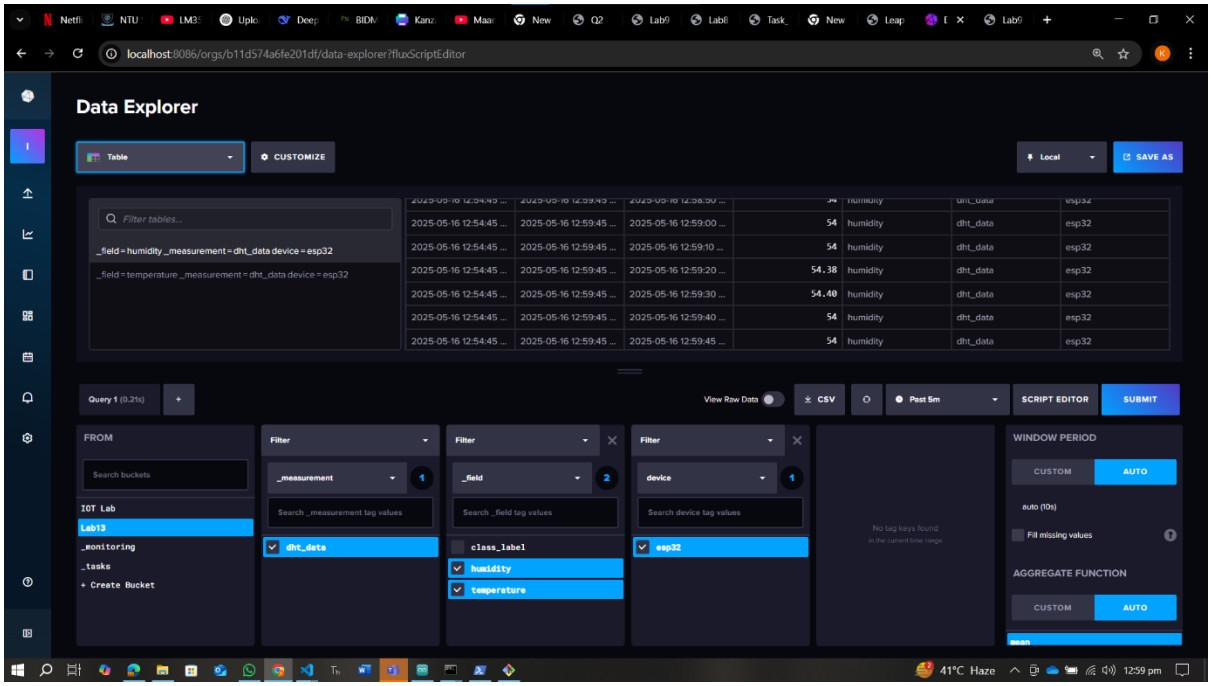
Task 2:

Execute the Python script 1-dht_data_only.py to **store MQTT data in InfluxDB**.

Now by running python code, it saves MQTT data to InfluxDB.



Here the data on the Dashboard of Influx DB is going successfully.

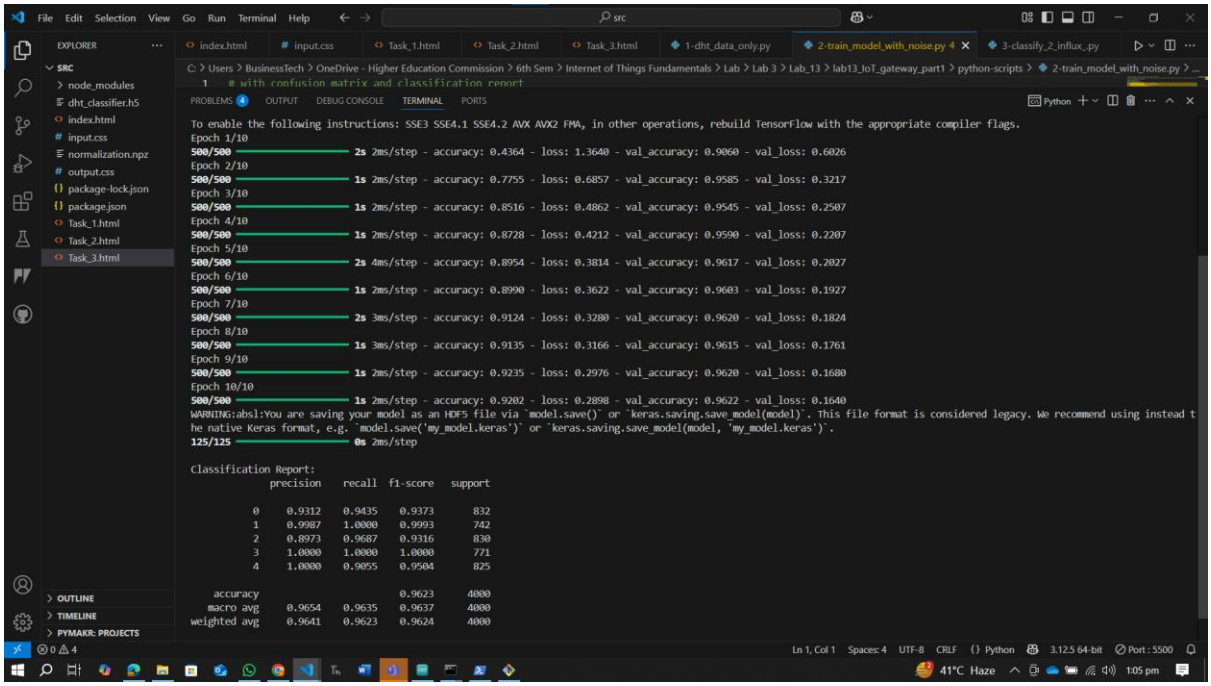


Task 3:

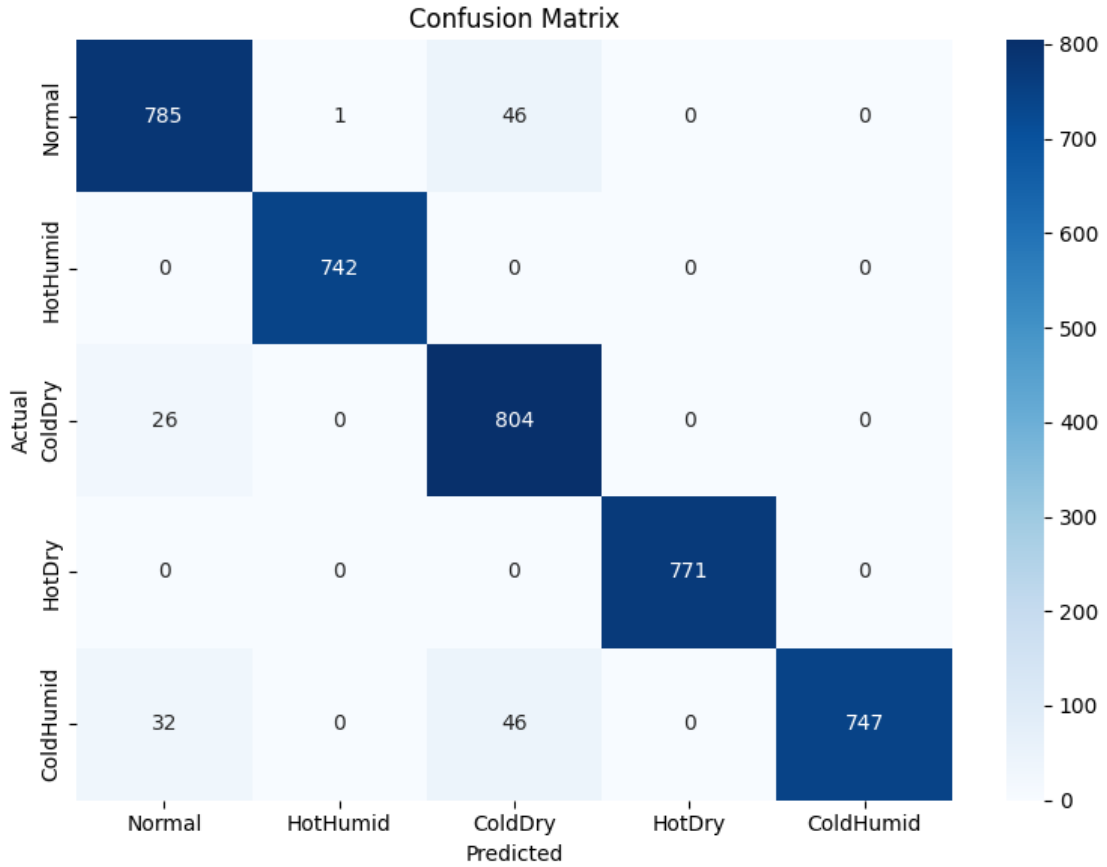
Run 2-train_model_with_noise.py and **record the confusion matrix and classification report**.

Now I train our model by running that code to create a synthetic dataset and

Create a neural network to classify temperature/humidity into 5 classes. And the model record confusion matrix and classification report.



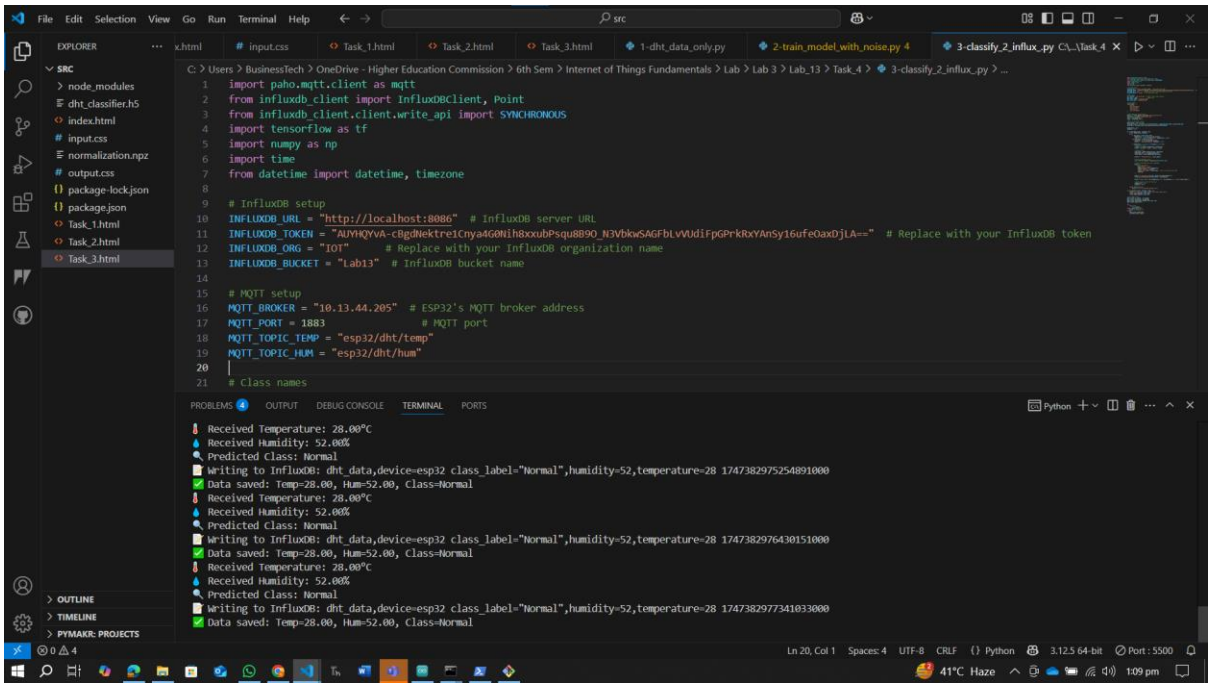
Here the Confusion Matrix of the model which I train. Showing the highest miss classification are done in Cold Humidity Class then Normal Class and so on.



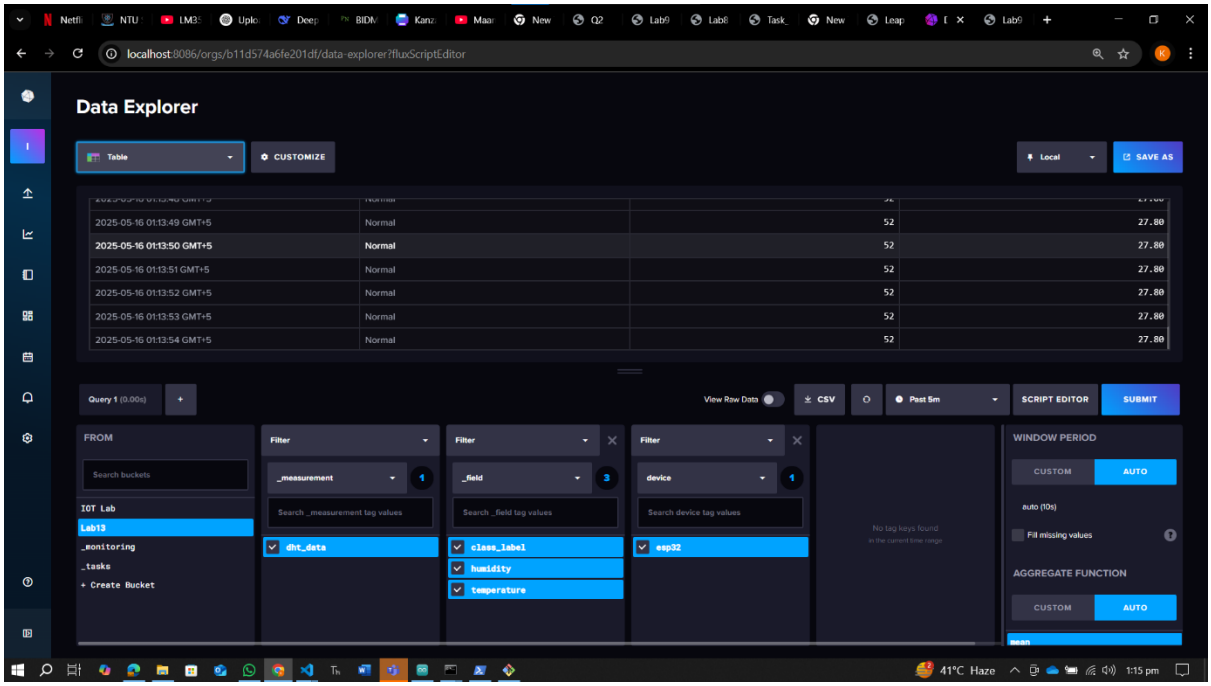
Task 4:

Execute 3-classify_2_influx.py and **verify InfluxDB data** for temperature, humidity, and classification results.

Now I run this data to read live sensor data, classify it using the trained model, and save that data. The terminal is showing real time data.



Here I check the data on the Dashboard of Influx DB is going successfully. Here the class_label is added as it shows the lable of the class as I used the train model to predict the class.


Data Explorer
Table
CUSTOMIZE
Local
SAVE AS
2025-05-16 01:13:49 GMT+5 Normal 52 27.80
2025-05-16 01:13:50 GMT+5 Normal 52 27.80
2025-05-16 01:13:51 GMT+5 Normal 52 27.80
2025-05-16 01:13:52 GMT+5 Normal 52 27.80
2025-05-16 01:13:53 GMT+5 Normal 52 27.80
2025-05-16 01:13:54 GMT+5 Normal 52 27.80
Query 1 (0.00s)
View Raw Data
CSV
Past 5m
SCRIPT EDITOR
SUBMIT
FROM
Search buckets
IOT Lab
Lab13
_monitoring
_tasks
+ Create Bucket
Filter
_measurement
Search _measurement tag values
dht_data
Filter
_field
Search _field tag values
class_label
humidity
temperature
Filter
device
Search device tag values
esp32
No tag keys found
in the current time range
WINDOW PERIOD
CUSTOM
AUTO
auto (10s)
Fill missing values
AGGREGATE FUNCTION
CUSTOM
AUTO
41°C Haze
1:15 pm